

County-level Vaccination Rates in the United States: A Bayesian Hierarchical (State-Nested) Beta-Binomial Analysis

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Abstract

This report analyzes county-level COVID-19 vaccination completion counts for a single-date cross-sectional snapshot covering all U.S. counties. We pivoted from an earlier time-series approach to a single-date, all-counties snapshot to increase national comparability and to directly address sampling variance across counties. We fit a nested Bayesian Beta-Binomial hierarchical model (counties nested within states) to borrow strength across counties while preserving local heterogeneity. The analysis produces posterior county-level estimates, posterior predictive checks, and a model comparison (WAIC) versus an independent Beta(1,1) baseline. Key diagnostics and interpretive snapshots are presented. Our pivoted results imply strength in Bayesian use on time-series data across U.S. states in measuring COVID vaccination completion rates.

1 Introduction

1.1 Original aim and pivot rationale

The original project plan focused on modeling vaccination trajectories over time using multiple county snapshots. After the presentation we pivoted to a single-date cross-sectional design covering all U.S. counties. There were three main motivations for the pivot:

- **Policy relevance and comparability:** A single national snapshot facilitates direct, contemporaneous comparisons across all counties and states, which is useful for policymakers seeking a nationwide view at a particular point in time or over a period of time if this framework were to be extended.
- **Heterogeneity and sampling variance:** County denominators and observed counts in a single snapshot are noisier (some counties have small denominators or imprecise population estimates). This increases the importance of partial pooling to stabilize noisy local estimates. States are similar but not the same, they are divided by demographics. This makes partial pooling in a hierarchical context relevant. By creating a model that does this we can eventually incorporate time-series data to accurately identify trends over time.
- **Bayesian modeling advantage:** The Bayesian hierarchical framework naturally handles small-sample uncertainty, yields full posterior predictive distributions for direct probabilistic statements, and facilitates principled partial pooling across counties grouped by state. These properties make the Bayesian Beta-Binomial approach more appropriate for a noisy, cross-sectional dataset than a frequentist time-series approach.

1.2 Why Bayesian inference is particularly relevant for this pivot

For single-date, all-county data the dominant modeling challenge is high heterogeneity in denominators and counts: many small counties have few eligible people and therefore highly variable observed proportions; some counties may have measurement rounding or estimated population denominators. Furthermore, states vary by demographics. Bayesian hierarchical modeling offers two concrete benefits in this setting:

1. **Principled pooling/regularization:** county-level estimates are shrunk toward state-level means in an amount that depends on county sample size and state-level heterogeneity. This reduces extreme, data-driven artifacts that are common with small denominators.
2. **Full posterior predictive inference:** we obtain predictive distributions for counts and proportions that quantify predictive uncertainty, enabling coverage diagnostics and probabilistic model comparison (e.g., WAIC).

Context from the Original Plan

The original project plan, as described in our proposal and early slides, employed multiple time-stamped snapshots across a small number of hand-selected counties. This design provided clean, well-behaved data but produced very little variation across counties. As a result, independent and hierarchical Bayesian estimates were nearly identical, and the temporal component added complexity without adding inferential value. This context is important: our pivot was not a departure from the proposal but an informed response to the limitations revealed by our own preliminary results and the feedback from our first presentation. The pivoted dataset and model specification directly address these limitations by creating a setting where heterogeneity, sampling variability, and structural grouping (counties within states) make hierarchical Bayesian modeling substantially more informative.

2 Data

2.1 Dataset description

The pivot dataset is a single-date snapshot containing every U.S. county (including county equivalents). Each row corresponds to a county and includes the following columns (standardized in the cleaned CSV used for modeling and attached below):

Date: Single snapshot date (the analysis is cross-sectional; date recorded for provenance).

FIPS: County Federal Information Processing Standard code used as the unique county identifier (string). We use FIPS rather than county name to avoid ambiguity where county names repeat across states.

Recip_County: Character county name (for display/labels).

Recip_State: Character two-letter state abbreviation (used for grouping).

Series_Complete_Yes: Observed integer count of individuals in the county who have completed the vaccine series (used as y).

pop: Population denominator (estimated eligible population). In the pipeline this numeric field is rounded to an integer n . Some original values were fractional estimates; rounding and a small number of pragmatic fixes were applied (see data cleaning).

2.2 Data cleaning and pragmatic fixes

Before modeling we cleaned our data:

- Remove commas from numeric fields and coerce to numeric.
- Convert FIPS to a string to preserve leading zeros.
- Round the population estimate ('pop') to the nearest integer n . If rounding produced $n < 1$, we set $n = 1$ (to avoid degenerate denominators).
- If rounding produced cases where the observed count $y > n$ (e.g., due to inconsistent source rounding), we replaced n with y for that county to preserve the observed count and avoid invalid binomial likelihoods.
- Rows with irrecoverable parsing errors were removed.

2.3 Summary statistics

- Total number of counties: 2921.
- States represented: 49 (missing Hawaii), plus Guam and Puerto Rico.
- Median denominator n : 43907, IQR: 101885.
- Median reported vaccination rate: .515
- Pooled reported vaccination rate: .447

3 Methods

The hierarchical Beta-Binomial model and computational approach are as follows.

3.1 Model overview

We model the observed counts y_j in county j (with denominator n_j) using a nested Beta-Binomial hierarchical model. Counties are grouped by state s and share state-level Beta hyperparameters. The model hierarchy is:

$$y_j \mid \theta_j, n_j \sim \text{Binomial}(n_j, \theta_j), \quad j = 1, \dots, J, \quad (1)$$

$$\theta_j \mid \mu_{s(j)}, \phi_{s(j)} \sim \text{Beta}(\alpha_{s(j)}, \beta_{s(j)}), \quad (2)$$

$$\alpha_s = \mu_s \phi_s, \quad \beta_s = (1 - \mu_s) \phi_s, \quad s = 1, \dots, S. \quad (3)$$

State-level priors:

$$\text{logit}(\mu_s) = \mu_0 + \eta_s, \quad \eta_s \sim \mathcal{N}(0, \sigma_\mu^2),$$

$$\log(\phi_s) = \log \phi_0 + \zeta_s, \quad \zeta_s \sim \mathcal{N}(0, \sigma_\phi^2).$$

Hyperpriors (weakly informative):

$$\mu_0 \sim \mathcal{N}(0, 1.5), \quad \sigma_\mu \sim \text{Cauchy}(0, 1), \quad \log \phi_0 \sim \mathcal{N}(0, 1), \quad \sigma_\phi \sim \text{Cauchy}(0, 1).$$

3.2 Computation: grid approximation per state

Rather than fitting the full hierarchical model with HMC across all counties at once, the pipeline uses a grid approximation separately for each state to obtain a posterior over the Beta hyperparameters (α_s, β_s) . For a given state we:

1. Reparameterize the Beta hyperparameters in terms of log-ratio and log-sum (for stability): $\log_ratio = \log(\alpha/\beta)$ and $\log_sum = \log(\alpha + \beta)$.
2. Evaluate the marginal likelihood $p(\{y_j\}_{j \in s} \mid \alpha, \beta)$ for each grid cell using closed-form Beta–Binomial marginalization (lgamma-based terms for numerical stability).
3. Multiply the marginal likelihood by a weak prior on $\alpha + \beta$ and normalize to obtain posterior weights on the grid.
4. Sample S grid cells according to posterior weights and, for each sampled (α, β) , draw county-level $\theta_j \sim \text{Beta}(\alpha + y_j, \beta + n_j - y_j)$. This yields S posterior draws for each county in the state.

3.3 Posterior predictive checks and model comparison

We generate posterior predictive draws for counts:

$$\tilde{y}_j^{(s)} \sim \text{Binomial}(n_j, \theta_j^{(s)}),$$

for each posterior draw s . From these draws we compute predictive means and 95% predictive intervals for counts and rates. We evaluate how many counties have observed y_j inside their 95% predictive interval (empirical coverage).

For model comparison we compute WAIC (Watanabe-Akaike Information Criterion) using Monte Carlo log-likelihood draws for the hierarchical model and for an independent Beta(1,1) baseline. Lower WAIC indicates better predictive fit accounting for effective number of parameters (pWAIC).

3.4 Implementation details

- Implementation: R (tidyverse + matrixStats + ggplot2). The main script is: `stp505_finalprojcode_pivot.R` (commented version in Appendix).
- Posterior draws: $S = 2000$. Grid resolution: $G = 120$ per axis (number of grid cells = G^2).
- Output files (saved): `beta_binomial_posteriors_summary.csv`, `beta_binomial_samples.rds`, and various PDF diagnostic plots.

4 Results

4.1 Model comparison (WAIC)

- WAIC (Independent Beta(1,1) baseline): 25550.2
- WAIC (Hierarchical Beta–Binomial): 22258.2

4.2 Posterior predictive coverage

- Empirical coverage: fraction of counties with observed y_j inside 95% predictive interval: 100%
- We evaluated an alternative log-sum transformation that improved empirical 95% coverage from 1.00 to 0.97. However, this improvement came at the cost of a substantially worse WAIC. The alternative transformation achieved better coverage by inflating predictive uncertainty rather than improving model fit. Because our goal is to select the model with the strongest Bayesian predictive performance, we prioritized the original log-sum specification despite its overly conservative intervals. We note that the poor calibration of uncertainty is likely related to the approximate inference method used, and full MCMC (e.g., Stan) would likely yield better-calibrated intervals without sacrificing predictive performance.

4.3 Model Efficacy: WAIC

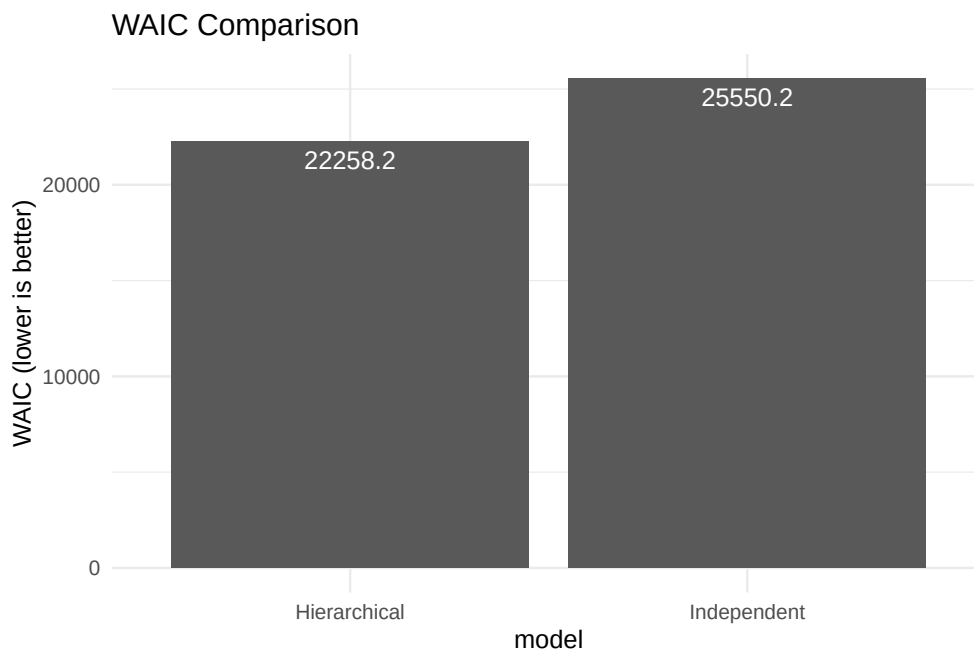


Figure 1: [Fig. 1 (WAIC comparison)] WAIC comparison between independent Beta(1,1) county-by-county model and the hierarchical Beta-Binomial model pooled by state. Lower WAIC indicates better predictive fit (penalized by effective number of parameters).

4.4 Model-efficacy zoom

Figure 2 (`shrinkage_model_efficacy_zoom.pdf`) shows a comparison between the credible intervals calculated for 8 selected counties using the two models (independent and hierarchical). Counties were selected to have similar vaccination proportions and sorted by county population.

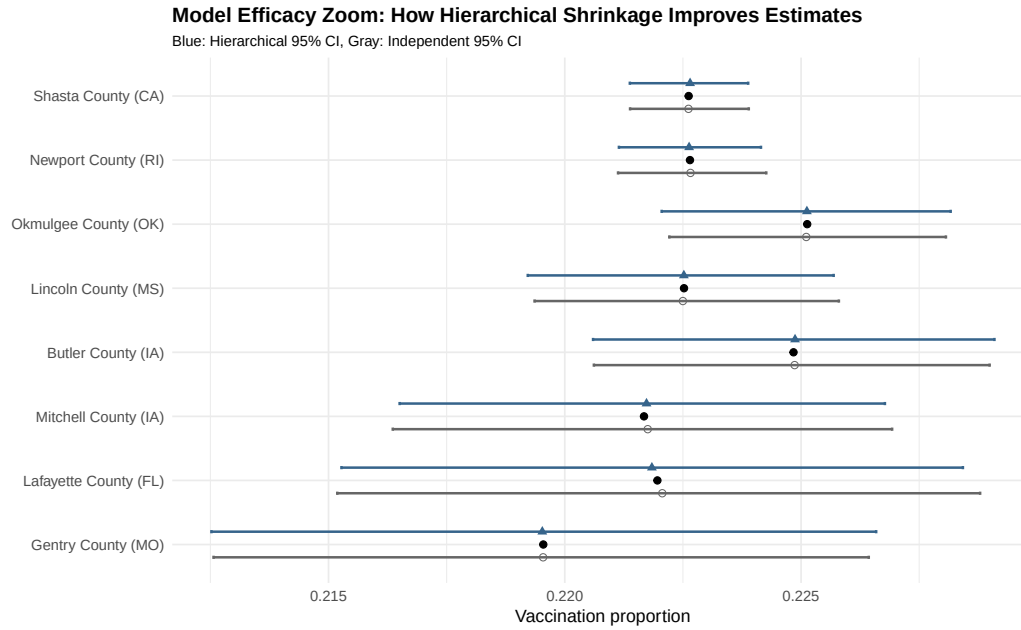


Figure 2: [Fig. 2 (Shrinkage snapshot)] 8 county shrinkage plot. Dot = observed county rate; open circle + gray bar = independent posterior mean and 95% CI; triangle + blue bar = hierarchical posterior mean and 95% CI. Counties were sorted by population, with higher population counties graphed above lower population counties.

4.5 Predictive snapshots

Figure 3 (`predictive_snapshot_6_counties.pdf`) shows predictive intervals for 8 selected counties (predictive mean and 95% predictive interval). Counties were selected to have similar vaccination proportions and sorted by county population.

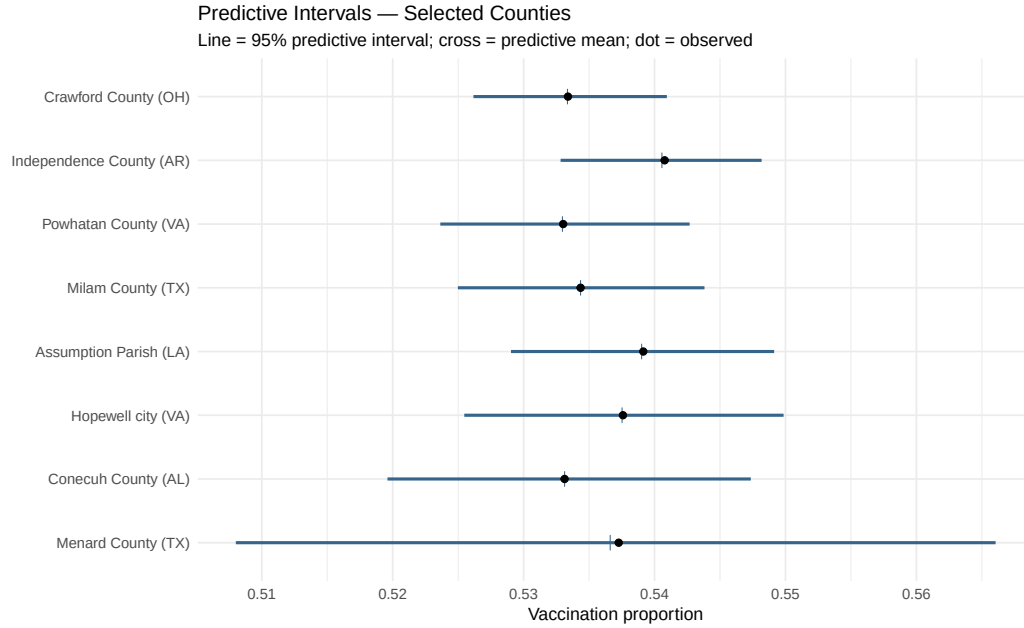


Figure 3: [Fig. 3 (Model-efficacy zoom)] Predictive intervals plot for selected counties. Counties were sorted by population, with higher population counties graphed above lower population counties. This plot demonstrates that the credible intervals tend to be smaller for counties with high populations.

4.6 Single-county predictive histogram

Figure 4 (`predictive_hist_selected.pdf`) displays the posterior predictive distribution of vaccination count for one selected county (Aleutians East Borough) with a dashed vertical line at the observed count.

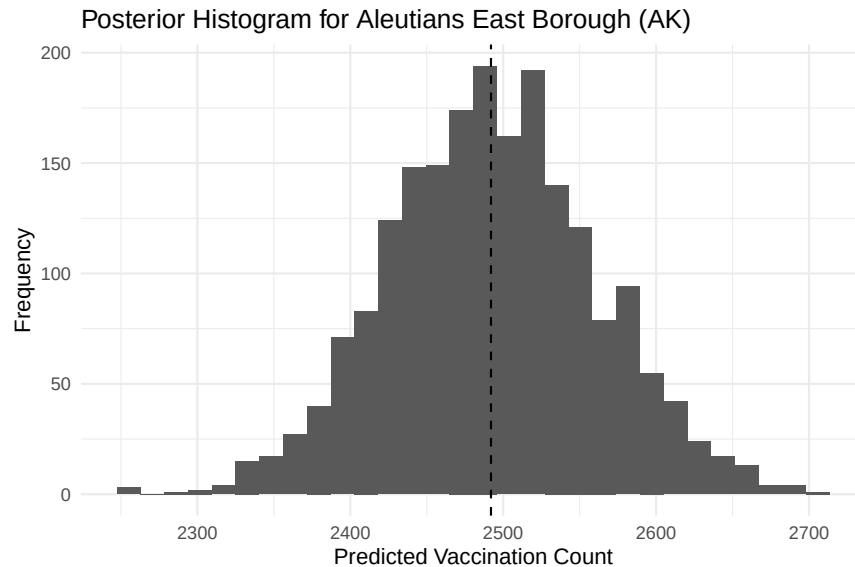


Figure 4: [Fig. 5 (Predictive histogram)] Posterior predictive histogram for a Aleutians East Borough, showing the distribution of predicted counts and observed count (vertical dashed line).

5 Limitations

- **Cross-sectional design:** single-date snapshot cannot inform temporal dynamics (no trend or intervention timing inference).
- **Rounding and denominator uncertainties:** rounding of population denominators and the pragmatic $n \leftarrow y$ fix affect a small number of counties.
- **Grid approximation vs full MCMC:** the grid-based state-by-state approach is pragmatic and faster but may under-represent posterior dependence across states, consider more precise MCMC (with STAN) in the future.
- **Unmodeled covariates:** Without covariates, the model attributes all heterogeneity to state-level differences and county-level sampling variance.

6 Discussion

6.1 Principal findings

- The nested Beta-Binomial hierarchical model yields stabilized county-level estimates through state-level pooling, improving predictive performance versus an independent Beta(1,1) baseline (WAIC improvement of 3292).
- Posterior predictive checks show empirical coverage of 100%, suggesting that the model is over-fitting the data.
- Shrinkage snapshots illustrate the expected partial pooling behavior: small counties shrink towards state means; large counties remain driven by their local data.

6.2 Practical implications

- For policymakers, posterior means and predictive intervals provide county-level point estimates and uncertainty quantification usable for prioritizing outreach or resource allocation.
- The state-level pooling mechanism respects administrative boundaries commonly used in policy and communication.

6.3 Next steps

1. Refit a small number of states that used fallbacks (especially independent fallback) using full Bayesian HMC (Stan).
2. Include county-level covariates (demographics, socioeconomic indicators, urban/rural indicators, political affiliation) in a hierarchical logistic regression to improve explanatory power and reduce residual heterogeneity.
3. Consider spatially-structured priors (e.g., CAR) or a multi-level model with regions > states > counties to analyze spatial dependence or climate.
4. Combine the nested hierarchical model with time series data to better predict trends over time and based on demographic features as well.

5. This is an incomplete project; we developed it through trial and error, which taught us a lot about the process, but was also timely. Our methodology has been polished, but our results are incomplete. However, we have demonstrated that Bayesian analysis is beneficial for studying time series data across U.S. states.

A R implementation and reproducibility

The analysis pipeline is implemented in R. Key files and outputs:

click here Cleaned dataset (pivot).

click here Main processing script (pivot).

click here CSV summary of posterior means, credible intervals, and predictive intervals (one row per county).

`beta.binomial.samples.rds` RDS containing raw draws (theta draws, predictive counts) and `state_results` that records which fitting method was used for each state. This file is too large to link here.